

IEAP RStudio Series 03

Data Wrangling and ANOVA

Jennifer HOUMARD

Valentin BALBINE

Shakir NASIR

12 October 2026

Table of contents

1	ANOVA: Analysis of Variance	2
1.1	Think before coding	2
1.1.1	Scientific question	2
1.1.2	How to visualize the effect of difficulty?	2
1.1.3	How to test whether there is a significant difference?	2
1.1.4	How to report the results?	2
1.2	What libraries are required to answer the above questions?	2
1.2.1	Packages used	2
1.3	Data reorganization	2
1.3.1	Data preparation	2
1.3.2	ANOVA analysis	6
1.4	Run the linear regression by GROUP	9
1.4.1	Regression equations	9
1.5	Save a graph in a PDF file for publication	10
1.5.1	Create the publication figure	10
1.6	Export the graph as a PDF file for publication	10
1.7	Discuss the differences with the article	10
1.7.1	Comparison with the article	10
1.7.2	Conclusion	10
2	Git Workflow Strategy	10
3	Challenges and Lessons Learned	10
4	Data Sources	10
5	Scientific References	10
6	Grading Checklist	10

```
library(tidyverse)
library(ez)
library(rstatix)
```

GitHub repository: <https://github.com/jenniferhoumard/IEAP-Series03-RStudio>

1 ANOVA: Analysis of Variance

1.1 Think before coding

1.1.1 Scientific question

1.1.2 How to visualize the effect of difficulty?

1.1.3 How to test whether there is a significant difference?

1.1.4 How to report the results?

1.2 What libraries are required to answer the above questions?

1.2.1 Packages used

1.3 Data reorganization

1.3.1 Data preparation

Before running the statistical analysis, the dataset was checked to make sure the data were complete and organized correctly.

The experiment originally included:

- three groups: ap, cp, and pp
- three ID conditions: ID1, ID2, and ID3
- up to three repetitions: R1, R2, and R3

However, not all participants completed every ID condition and every repetition. Therefore, the dataset had to be checked for missing or incomplete conditions before creating the final dataset used for the analysis.

The groups are:

- ap = healthy adult group

- cp = healthy young group
- pp = stroke group

1.3.1.1 Import and inspect the data

The raw dataset was first imported and inspected to understand its structure, variables, and dimensions.

```
data_raw <- read.table("data/Results.csv", header = TRUE)
```

```
head(data_raw) # display data
```

	GROUP	SUBJ	TRIAL	ORI	DIR	ID	REP	NbVelPeaks	MovementTime
1	ap	apS01	T01	W	outward	ID1	R1	2	1.23
2	ap	apS01	T01	W	inward	ID1	R1	2	1.26
3	ap	apS01	T01	NW	outward	ID1	R1	3	1.14
4	ap	apS01	T01	NW	inward	ID1	R1	2	0.90
5	ap	apS01	T01	N	outward	ID1	R1	3	1.11
6	ap	apS01	T01	N	inward	ID1	R1	3	0.81

```
str(data_raw) # structure of data
```

```
'data.frame': 7560 obs. of 9 variables:
 $ GROUP      : chr  "ap" "ap" "ap" "ap" ...
 $ SUBJ       : chr  "apS01" "apS01" "apS01" "apS01" ...
 $ TRIAL      : chr  "T01" "T01" "T01" "T01" ...
 $ ORI        : chr  "W" "W" "NW" "NW" ...
 $ DIR        : chr  "outward" "inward" "outward" "inward" ...
 $ ID         : chr  "ID1" "ID1" "ID1" "ID1" ...
 $ REP        : chr  "R1" "R1" "R1" "R1" ...
 $ NbVelPeaks : num  2 2 3 2 3 3 1 2 4 1 ...
 $ MovementTime: num  1.23 1.26 1.14 0.9 1.11 0.81 1.08 0.81 1.32 1.02 ...
```

```
dim(data_raw) # dimension of data (rows, columns)
```

```
[1] 7560    9
```

1.3.1.2 Check GROUP × ID completeness

Next, the number of observations/rows for each combination of GROUP and ID:

```
data_raw %>%  
  count(GROUP, ID)
```

	GROUP	ID	n
1	ap	ID1	740
2	ap	ID2	740
3	ap	ID3	740
4	cp	ID1	1140
5	cp	ID2	1140
6	cp	ID3	1140
7	pp	ID1	760
8	pp	ID2	760
9	pp	ID3	400

For the **ap** group, ID1, ID2, and ID3 each contain 740 observations. For the **cp** group, all three IDs each contain 1140 observations. This shows that the three ID conditions are complete within these two groups.

For the stroke group (**pp**), ID1 and ID2 each contain 760 observations, while ID3 contains only 400 observations. This shows that ID3 is incomplete for the **pp** group because some stroke participants did not complete this condition.

Therefore, ID3 will be removed before the ANOVA analysis so that the same ID conditions are available across all three groups.

1.3.1.3 Remove the incomplete ID3 condition

The previous check showed that ID3 was incomplete for the stroke group (**pp**). Therefore, all observations belonging to ID3 were removed before continuing the analysis.

```
data_clean <- data_raw %>%  
  filter(ID != "ID3")  
  
head(data_clean)
```

	GROUP	SUBJ	TRIAL	ORI	DIR	ID	REP	NbVelPeaks	MovementTime
1	ap	apS01	T01	W	outward	ID1	R1	2	1.23
2	ap	apS01	T01	W	inward	ID1	R1	2	1.26

3	ap	apS01	T01	NW	outward	ID1	R1	3	1.14
4	ap	apS01	T01	NW	inward	ID1	R1	2	0.90
5	ap	apS01	T01	N	outward	ID1	R1	3	1.11
6	ap	apS01	T01	N	inward	ID1	R1	3	0.81

1.3.1.4 Check repetitions

```
data_clean %>% count(GROUP, REP)
```

	GROUP	REP	n
1	ap	R1	520
2	ap	R2	520
3	ap	R3	440
4	cp	R1	760
5	cp	R2	760
6	cp	R3	760
7	pp	R1	760
8	pp	R2	760

After removing ID3, ID1 and ID2 contain the same number of observations within each group.

The total number of observations differs between groups because the groups contain different numbers of participants. This does not mean that the ID data are incomplete.

The next step is therefore to check whether the repetitions (REP) are complete within each group.

1.3.1.5 Remove the incomplete R3 repetition

The previous check showed that the third repetition (R3) was not completed consistently across all groups. Therefore, R3 was removed from the dataset.

```
data_final <- data_clean %>% filter(REP != "R3") #remove all rows belonging to R3
head(data_final)
```

	GROUP	SUBJ	TRIAL	ORI	DIR	ID	REP	NbVelPeaks	MovementTime
1	ap	apS01	T01	W	outward	ID1	R1	2	1.23
2	ap	apS01	T01	W	inward	ID1	R1	2	1.26
3	ap	apS01	T01	NW	outward	ID1	R1	3	1.14
4	ap	apS01	T01	NW	inward	ID1	R1	2	0.90
5	ap	apS01	T01	N	outward	ID1	R1	3	1.11
6	ap	apS01	T01	N	inward	ID1	R1	3	0.81

The number of observations was then checked again for each group and repetition:

```
data_final %>% count(GROUP, REP)
```

	GROUP	REP	n
1	ap	R1	520
2	ap	R2	520
3	cp	R1	760
4	cp	R2	760
5	pp	R1	760
6	pp	R2	760

1.3.1.6 Check participants by GROUP

Finally, the number of participants in each group was checked.

```
data_final %>% distinct(GROUP, SUBJ) %>% count(GROUP)
```

	GROUP	n
1	ap	13
2	cp	19
3	pp	19

1.3.1.7 Create the cleaned dataset

The final cleaned dataset is `data_final`.

It contains:

- all three groups: `ap`, `cp`, and `pp`
- two complete ID conditions: `ID1` and `ID2`
- two complete repetitions: `R1` and `R2`

`ID3` was removed because it was incomplete for the stroke group (`pp`), and `R3` was removed because it was not completed consistently across all groups.

The cleaned dataset is now ready for the statistical analysis.

1.3.2 ANOVA analysis

Now that we have explored and filtered the data, we can proceed with the ANOVA analysis.

1.3.2.1 Prepare the data for ANOVA

To avoid giving excessive weight to repeated trials (at least 2 for each direction as indicated in the experimental design), movement time was averaged for each participant (SUBJ), group (GROUP) and index of difficulty (ID). The resulting dataset contained one mean movement time value per participant and difficulty level and was used for the subsequent statistical analysis.

1.3.2.2 Run the ANOVA

```
S <- data_final|>
  group_by(GROUP, SUBJ, ID) |>
  summarise(MT = mean(MovementTime), .groups = "drop")
res <- anova_test(data = S, dv = MT, wid = SUBJ,
                  between = GROUP, within = ID)
anova_tab <- get_anova_table(res)

anova_tab
```

ANOVA Table (type III tests)

	Effect	DFn	DFd	F	p	p<.05	ges
1	GROUP	2	48	37.244	1.72e-10	*	0.575
2	ID	1	48	26.109	5.54e-06	*	0.066
3	GROUP:ID	2	48	11.581	7.87e-05	*	0.059

1.3.2.3 Interpret the ANOVA results

Group effect : The 3 groups (ap, cp, pp) were compared to each other. The ANOVA results showed a significant effect of GROUP on movement time $F(2,48) = 37.24$, $p < 0.001$. This indicates that the mean movement time differed significantly between the three groups.

ID effect : The two levels of ID (ID1, ID2) were compared to each other. The ANOVA results showed a significant effect of ID on movement time ($F(1,48) = 26.11$, $p < 0.001$). This indicates that the mean movement time differed significantly between the two levels of ID. When the ID increased, the movement time increased as well.

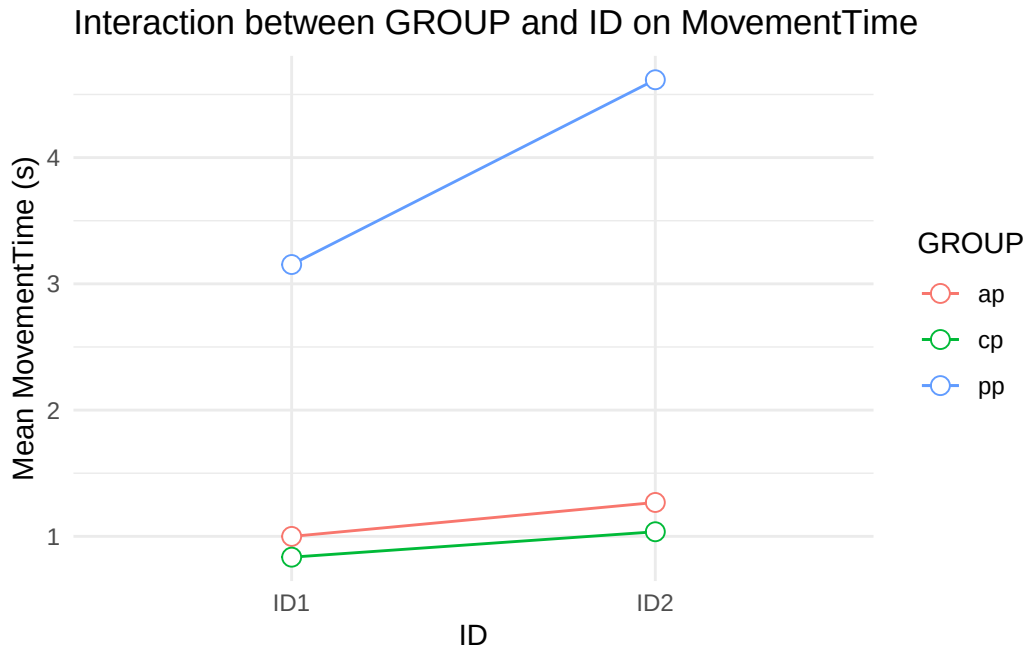
Crossed effect (GROUP $\times$ ID) : The interaction between GROUP and ID was the most significant ($F(2,48) = 11.58$, $p < 0.001$). This indicates that the effect of ID on movement time differed between the three groups and was way more important on the stroke group (pp). In other words, the difference in mean movement time between ID1 and ID2 was not the same for all groups.

In conclusion, the effect of difficulty is not the same for all groups, and the effect of group is not the same for all levels of difficulty. This suggests that the relationship between movement time and difficulty level is influenced by the group to which participants belong.

1.3.2.4 Plot the GROUP × ID interaction

```
ggplot(data_final, aes(x = ID, y = MovementTime, color = GROUP)) +
  stat_summary(fun = mean, geom = "line", aes(group = GROUP)) +
  stat_summary(fun.data = mean_sdl, fun.args = list(mult = .5),
    geom = "errorbar", width = 0.01) +
  stat_summary(fun = mean, geom = "point", shape = 21, fill = "white", size = 3) +
  labs(title = "Interaction between GROUP and ID on MovementTime",
    x = "ID", y = "Mean MovementTime (s)") +
  theme_minimal()
```

Warning: Computation failed in `stat\_summary()`.
 Caused by error in `fun.data()`:
 ! The package "Hmisc" is required.



Graphically, we can see that a significant GROUP × ID interaction was observed, $F(2,48) = 11.58$, $p < 0.001$. As shown in the graph displayed above, movement time increased with task difficulty in all groups, but the increase was larger for the stroke group (PP) than for the healthy groups (AP and CP).

1.4 Run the linear regression by GROUP

The aim is to see how MovementTime changes when the difficulty (ID) increases, and whether this change is different between the three groups.

A separate linear regression is calculated for each group:

ap = healthy adults

cp = healthy young adults

pp = stroke group

The regression equation is:

$$y = mx + b$$

$$MT = a + b(ID)$$

where:

MT = Movement Time

a = intercept

b = slope

ID = Index of Difficulty

The slope tells us how much Movement Time changes when ID increases.

1.4.1 Regression equations

For the ANOVA, ID was treated as a categorical variable with two conditions: ID1 and ID2.

For the linear regression, we need to convert the categorical ID variable to a numerical variable with the actual Index of Difficulty values:

- ID1 = 3.80
- ID2 = 5.43

A new variable called ID_numeric is therefore created.

```
regression_data <- S %>%  
  mutate(ID_numeric = case_when(ID == "ID1" ~ 3.80, ID == "ID2" ~ 5.43))
```

1.5 Save a graph in a PDF file for publication

1.5.1 Create the publication figure

1.6 Export the graph as a PDF file for publication

1.7 Discuss the differences with the article

1.7.1 Comparison with the article

1.7.2 Conclusion

2 Git Workflow Strategy

3 Challenges and Lessons Learned

Once more the challenges was to cross our different code by checking the variable names and the data structure. We also had to check the data for missing values and outliers, which was a bit tricky. But we learned a lot about data cleaning and manipulation, as well as how to perform ANOVA analysis in R. We also learned how to interpret the results and create visualizations to present our findings.

4 Data Sources

5 Scientific References

6 Grading Checklist